Fiscal Year:	FY 2020	**Task Last Updated:**	FY 10/23/2019
PI Name:	Olson, Sandra Ph.D.		
Project Title:	Correlation of Normal Gravity Aerospace Materials Flammability Data with Flammability in Microgravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Glenn Research Center		
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Zip Code:	44135	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2010	**End Date:**	12/31/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Urban, David	**Contact Phone:**	216-433-2835
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Flight Program:			
Flight Assignment:	NOTE: Changed end date to 12/31/2022, per PI (Ed., 10/3/19) NOTE: Extended to 12/31/2019, per PI (Ed., 1/26/2015) NOTE: Extended to 12/31/2014, per PI (Ed., 10/17/13) NOTE: Extended to 12/31/2013, per PI (Ed., 10/10/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hirsch, David M.S. (NTEC, Jacobs Engineering)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA STD 6001 Test 1 is the major method used to evaluate flammability of materials intended for use in habitable environments of U.S. spacecraft. The method is an upward flame propagation test initiated in a stagnant environment and using a well-defined igniter flame at the bottom of a vertically mounted sample. A material passes this test if the vertical burn length is less than 15.2 cm (6 inches) and there is no evidence of transfer of burning debris. The test is conducted in the most severe flaming combustion environment (oxygen concentration, pressure) expected in the spacecraft, currently 30% oxygen at 10.2 psia. Recent research has shown that current normal gravity materials flammability tests do not correlate with flammability in ventilated, micro-, or reduced-gravity conditions. The materials selection for spacecraft is based on the assumption of commonality between ground test flammability results and spacecraft environments, which does not appear to be valid. To better understand the actual normal gravity flammability limits, a modified Test 1 protocol has been proposed that, as an alternative to qualifying materials as pass/fail in the worst-expected environments, measures the actual upward flammability limit (akin to the Limiting Oxygen Index (LOI)) for the material so that a more accurate assessment of the margin of safety of the material in the real use environment can be made. The method allows the option of selecting better or best materials based on the margin of safety, as opposed to what would be considered just "passing" from a flammability point of view. A research program is underway to correlate normal gravity flammability test data with data under ventilated reduced gravity or microgravity conditions. The program will determine materials suitability for microgravity and reduced gravity testing by evaluating oxygen concentration thresholds under various total pressures in normal gravity tests and in spacecraft-specific environments (ventilated microgravity or reduced partial gravity). The normal gravity data will be correlated with flammability data in microgravity environments and the safety factors involved when selecting spacecraft materials based on normal gravity testing will be evaluated. Rigorous correlations between ground test flammability data and data in ventilated spacecraft environments would allow selection of materials with increased fire resistance, and thus decrease the fire risk in space systems. The operational flexibility will also increase since the safety factors will be assessable.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The identification of the fundamental limit in microgravity and its comparison with the normal gravity flammability limit will help us understand flammability in different configurations on Earth. It will improve the safety of astronauts in spaceflight so they can return safely to Earth.
Task Progress:	Low-pressure blowoff experiments were conducted with a stagnation flame stabilized on the forward tip of cast polymethyl methacrylate (PMMA) rods in a vertical wind tunnel. Pressure, forced flow velocity, gravity, and ambient oxygen concentration were varied. Both normal gravity and microgravity tests were conducted to determine the influence of buoyant flow velocity on the blowoff limits. A linear shift of a buoyant flow of ~25 cm/s was adequate to match the microgravity data. The flame standoff distance before blowoff varies with the inverse square root of pressure and stretch rate via the gas phase length scale d = (a/a)1/2, where the thermal diffusivity varies inversely with pressure. For a constant critical Damkohler number, the reaction rate must increase in tandem with the flow rate. The flame standoff distance responds to environment changes to maintain both the critical Damkohler number and the molar flux of oxygen along the blowoff boundary. These interdependencies result in the partial pressure of oxygen at blowoff being linearly dependent on the total pressure regardless of forced flow velocity and gravity level. At very low pressures, reactant leakage appears to cause a two-stage reaction zone. For very dilute oxidizer mixtures, an apparent high pressure blowoff limit is found, believed to be caused by chain-terminating three body reactions. A normal gravity materials flammability screening methodology is suggested for spacecraft exploration atmospheres where the blowoff partial pressure of oxygen versus total pressure blowoff limits are measured for the expected exploration atmospheres (oxygen concentrations). As long as the blowoff boundary is above the Earth-normal partial pressure of oxygen (21 kPa) threshold, the material is safe for use in spacecraft.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Articles in Peer-reviewed Journals	Marcum JW, Ferkul PV, Olson SL. "PMMA rod stagnation region flame blowoff limits at various radii, oxygen concentrations, and mixed convection stretch rates." Proceedings of the Combustion Institute. 2019;37(3):4001-8. (The 37th meeting of the International Combustion Institute) https://doi.org/10.1016/j.proci.2018.05.081 , Jan-2019
Articles in Peer-reviewed Journals	Olson SL, Ferkul PV, Marcum JW. " High speed video analysis of flame oscillation along a PMMA rod after stagnation region blowoff." Proceedings of the Combustion Institute. 2019;37(2):1555-62. (The 37th meeting of the International Combustion Institute) https://doi.org/10.1016/j.proci.2018.05.080 , Jan-2019
Papers from Meeting Proceedings	Olson S, Ferkul P, Fernandez-Pello C, Miller F, Wichman I, Bhattacharjee S, T'ien J. "Flammability limits from BASS-II testing in microgravity compared to normal gravity limits." 49th International Conference on Environmental Systems, Boston, MA, July 7-11, 2019. 49th International Conference on Environmental Systems, Boston, MA, July 7-11, 2019. Paper ICES-2019-101. , Jul-2019